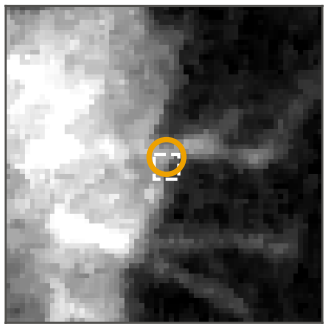
A 5 × 5 pixel patch of one test image (rows 336–340, columns 290–294); the full tensors are 512 × 512

x = input image
torch.Size([1, 1, 512, 512])



64 × 64 px crop; dashed box = 5 × 5 patch,
circle = outlined pixel

model
(UNet++)

logits = model(x)
torch.Size([1, 4, 512, 512])

[:, 0]
Background

-3.4	-3.9	-4.6	-5.0	-5.1
-2.0	-2.2	-2.6	-2.3	-1.9
-0.4	0.1	0.2	0.7	1.9
0.5	1.2	1.9	2.8	4.2
0.4	1.3	2.2	3.2	4.8

[:, 1]
F-actin

-5.0	-2.5	0.5	3.9	6.1
-4.8	-3.2	-1.0	1.0	2.3
-5.2	-4.4	-3.4	-2.8	-2.2
-5.4	-5.1	-4.8	-5.1	-5.4
-5.6	-5.3	-5.5	-6.2	-7.2

[:, 2]
Lamellipodia

-1.2	-0.4	-0.9	-1.5	-3.4
1.2	2.2	2.1	1.4	-0.3
3.6	4.5	4.3	3.6	2.1
5.4	6.3	5.9	4.9	3.2
6.2	7.0	6.7	5.4	3.5

[:, 3]
Focal adhesion

4.7	3.0	0.3	-3.4	-7.0
3.2	1.6	-0.5	-3.4	-5.8
1.5	0.0	-1.5	-3.4	-4.9
-0.1	-1.5	-2.7	-3.9	-5.0
-0.5	-2.0	-3.0	-3.8	-4.6

softmax
over dim=1

probs = torch.softmax(logits, dim=1)
torch.Size([1, 4, 512, 512])

0.00	0.00	0.00	0.00	0.00
0.00	0.01	0.01	0.01	0.01
0.02	0.01	0.02	0.06	0.45
0.01	0.01	0.02	0.11	0.74
0.00	0.00	0.01	0.10	0.78

0.00	0.00	0.50	0.99	1.00
0.00	0.00	0.04	0.41	0.92
0.00	0.00	0.00	0.00	0.01
0.00	0.00	0.00	0.00	0.00
0.00	0.00	0.00	0.00	0.00

0.00	0.03	0.12	0.00	0.00
0.12	0.62	0.89	0.57	0.07
0.87	0.98	0.98	0.94	0.54
0.99	0.99	0.98	0.89	0.26
1.00	1.00	0.99	0.90	0.22

1.00	0.96	0.38	0.00	0.00
0.88	0.37	0.06	0.00	0.00
0.11	0.01	0.00	0.00	0.00
0.00	0.00	0.00	0.00	0.00
0.00	0.00	0.00	0.00	0.00

argmax
over dim=1

pred = probs.argmax(dim=1)
torch.Size([1, 512, 512])

3	3	1	1	1
3	2	2	2	1
2	2	2	2	2
2	2	2	2	0
2	2	2	2	0

0 Background · 1 F-actin
2 Lamellipodia · 3 Focal adhesion

Outlined pixel: its 4 values along dim=1 are 0.00, 0.50, 0.12, 0.38 (sum = 1.00); argmax picks class 1 (F-actin). Every pixel is handled independently.